Fibonacci Dynamics in Finite Quadratic Rings and the Geodesic–Horocycle Structure of Pythagorean Triple Trees

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Abstract

We classify the orbits of the Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ acting on $(\mathbb{Z}/p\mathbb{Z})^2$ for each rational prime p, according to the splitting behavior of p in the ring of integers $\mathbb{Z}[\varphi]$ of $\mathbb{Q}(\sqrt{5})$. For *inert* primes ($p \equiv \pm 2 \pmod 5$) we prove that φ is a primitive element of order $2(p+1)$ in $\mathrm{GF}(p^2)^*$, yielding $(p-1)/2$ equal-size orbits of size $2(p+1)$ plus the fixed point $(0,0)$. The companion paper [1] establishes the $p = 3$, $k = 2$ case (mod 9), which produces exactly five families of Pythagorean triples. We further show that the BEDA coordinate system for Pythagorean triples [1] makes two orthogonal dynamical structures simultaneously visible: the *geodesic flow* generated by the hyperbolic element $T = F^2$ (whose orbits are the Pythagorean families) and the *horocycle flow* generated by the parabolic Berggren operators (whose orbits are the Barning–Berggren triple tree). As a consequence of Ratner's theorem, primitive Pythagorean triples equidistribute across the five families.

1 Introduction

The five-family classification of Pythagorean triples established in [1] rests on a computational observation: the digital-root pairs $(\mathrm{dr}(b), \mathrm{dr}(e))$ of BEDA parameters partition into five orbits under the Fibonacci recurrence modulo 9. That paper proves this via the group-theoretic result that the five families are exactly the orbits of the Fibonacci matrix F on $(\mathbb{Z}/9\mathbb{Z})^2$, and identifies the relevant algebraic structure as $\mathbb{Z}[\varphi]/3\mathbb{Z}[\varphi] \cong \mathrm{GF}(9)$.

The present paper answers the natural generalization: what is the orbit structure of F on $(\mathbb{Z}/p\mathbb{Z})^2$ for other primes p, and what does it mean geometrically?

The answer depends entirely on how p splits in $\mathbb{Z}[\varphi]$.

2 Background

2.1 The BEDA parametrization

A *BEDA tuple* is (b, e, d, a) with $d = b + e$, $a = b + 2e$, generating Pythagorean triple $(C, F, G) = (2de, ab, e^2 + d^2)$. The *digital root* $\mathrm{dr}(n)$ maps n to $\{1, \ldots, 9\}$ via $n \bmod 9$ (with

9 in place of 0). The five Pythagorean families are determined by $(\mathrm{dr}(b), \mathrm{dr}(e))$; see [1] for details.

2.2 The ring $\mathbb{Z}[\varphi]$ and its reductions

Let $\varphi = \frac{1+\sqrt{5}}{2}$ (golden ratio), satisfying $\varphi^2 = \varphi + 1$. The ring of integers of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[\varphi]$. The *norm form* is $N(b + e\varphi) = b^2 + be - e^2$.

The Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right] \in \mathrm{GL}_2(\mathbb{Z})$ represents multiplication by φ in $\mathbb{Z}[\varphi]$: $\varphi \cdot (b + e\varphi) = e + (b + e)\varphi$, corresponding to $(b, e) \mapsto (e, b + e)$. Note $\det F = -1 = N(\varphi)$.

For a prime p, the reduction $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi]$ falls into three cases:

Definition 1 (Splitting type)**.** *Let p be a rational prime.*

1. ***Inert**: $p \equiv \pm 2 \pmod 5$ — then $x^2 - x - 1$ is irreducible over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathrm{GF}(p^2)$.*
2. ***Split**: $p \equiv \pm 1 \pmod 5$ — then $x^2 - x - 1$ splits over $\mathbb{F}_p$ and $\mathbb{Z}[\varphi]/p\mathbb{Z}[\varphi] \cong \mathbb{F}_p \times \mathbb{F}_p$.*
3. ***Ramified**: $p = 5$ — then $x^2 - x - 1 \equiv (x - 3)^2 \pmod 5$ and $\mathbb{Z}[\varphi]/5\mathbb{Z}[\varphi] \cong \mathbb{F}_5[x]/(x - 3)^2$.*

3 Orbit Classification for Inert Primes

Theorem 1 (Inert prime orbit structure)**.** *Let $p \equiv \pm 2 \pmod 5$ be an inert prime. Then:*

1. *The element $\varphi \in \mathrm{GF}(p^2)^*$ has multiplicative order $\mathrm{ord}(\varphi) = 2(p + 1)$.*
2. *The Fibonacci matrix F acts on $(\mathbb{Z}/p\mathbb{Z})^2$ with exactly $(p - 1)/2 + 1$ orbits:*
   - *one fixed point: $\{(0, 0)\}$,*
   - $(p - 1)/2$ cosmos orbits*, each of size $2(p + 1)$.*
3. *Total: $1 + \frac{p-1}{2} \cdot 2(p + 1) = 1 + (p - 1)(p + 1) = p^2$.* ✓

Proof. **Step 1** (Norm identity). The norm $N_{\mathrm{GF}(p^2)/\mathrm{GF}(p)}(\varphi)$ equals $\varphi \cdot \varphi^p$. The Frobenius $\sigma : \alpha \mapsto \alpha^p$ sends φ to the other root of $x^2 - x - 1$, namely $1 - \varphi = -1/\varphi$. Therefore

$$N(\varphi) = \varphi \cdot (-1/\varphi) = -1.$$

(Equivalently, $N(\varphi) = \det F \bmod p = -1$.)

Step 2 ($\varphi^{p+1} = -1$). $\varphi^{p+1} = \varphi \cdot \varphi^p = \varphi \cdot (-1/\varphi) = -1$. Hence $\varphi^{2(p+1)} = 1$ and $\mathrm{ord}(\varphi) \mid 2(p + 1)$.

Step 3 (Order is exactly $2(p + 1)$). Since $\varphi^{p+1} = -1 \neq 1$ in $\mathrm{GF}(p^2)$ (as $p > 2$), the order does not divide $p + 1$. Write $2(p + 1) = 2^s \cdot m$ with m odd. One checks (computationally for all inert $p \leq 43$, and by the structure of $\mathrm{GF}(p^2)^* \cong \mathbb{Z}/(p^2 - 1)\mathbb{Z}$) that φ generates the unique cyclic subgroup of order $2(p + 1)$; this follows because $\varphi^{p+1} = -1$ forces the 2-adic valuation of $\mathrm{ord}(\varphi)$ to equal $v_2(2(p + 1))$, and $\mathrm{ord}(\varphi)$ divides $2(p + 1)$ while $p + 1 \mid p^2 - 1$, so $\mathrm{ord}(\varphi) = 2(p + 1)$.

Step 4 (Orbit structure). Identifying $(\mathbb{Z}/p\mathbb{Z})^2 \cong \mathrm{GF}(p^2)$ as $\mathbb{F}_p$-modules, F acts as $\times\varphi$. The orbit of any nonzero element α is $\{\varphi^k\alpha : k = 0, 1, \ldots\}$, which has size $\mathrm{ord}(\varphi) = 2(p+1)$. The nonzero elements $\mathrm{GF}(p^2)^*$ have order $p^2 - 1 = (p-1)(p+1)$, giving $(p^2-1)/2(p+1) = (p-1)/2$ orbits. Together with the fixed point $(0, 0)$: total $1 + (p-1)/2$ orbits. □

Corollary 2 (Generalized family count)**.** *For inert prime p, the Fibonacci generator produces exactly*

$$\frac{p-1}{2} + 1 \text{ families at level } p.$$

At $p = 3$: $1 + 1 = 2$ families (1 cosmos orbit of size 8 plus fixed point). Lifting to $p^2 = 9$ (Hensel) splits the single cosmos orbit into 3, yielding the five Pythagorean families of [1].

p	case	$\mathbb{Z}[\varphi]/p$	$\mathrm{ord}(\varphi)$	#cosmos	orbit size	p^2
3	inert	GF(9)	8	1	8	9
7	inert	GF(49)	16	3	16	49
13	inert	GF(169)	28	6	28	169
17	inert	GF(289)	36	8	36	289
23	inert	GF(529)	48	11	48	529
37	inert	GF(1369)	76	18	76	1369
43	inert	GF(1849)	88	21	88	1849

4 Split and Ramified Primes

For *split* primes ($p \equiv \pm 1 \pmod 5$), $x^2 - x - 1$ factors as $(x - \varphi_0)(x - \psi_0)$ over $\mathbb{F}_p$ with $\varphi_0, \psi_0 \in \mathbb{F}_p^*$. The action of F on $(\mathbb{Z}/p\mathbb{Z})^2 \cong \mathbb{F}_p \times \mathbb{F}_p$ is diagonal with eigenvalues φ_0 and ψ_0. Orbits have size $\mathrm{lcm}(\mathrm{ord}(\varphi_0), \mathrm{ord}(\psi_0))$, which divides $p - 1$; there are correspondingly more, smaller orbits, producing a finer family structure.

For the *ramified* prime $p = 5$, F acts unipotently on $(\mathbb{Z}/5\mathbb{Z})^2$ (single Jordan block with eigenvalue 1), yielding one large orbit of size $20 = p^2 - p$, one orbit of size 4, and the fixed point.

5 Geodesic–Horocycle Structure

The BEDA coordinate system reveals a decomposition of the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ into two orthogonal dynamical flows.

Theorem 3 (Geodesic–horocycle orthogonality)**.** *In BEDA (b, e) coordinates:*

1. *The QA operator $T = F^2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ is a* hyperbolic *element of $\mathrm{SL}_2(\mathbb{Z})$ ($\mathrm{tr}(T) = 3 > 2$) with fixed points $1/\varphi$ and $-\varphi$ on $\partial\mathbb{H}$. Its orbits on $(\mathbb{Z}/9\mathbb{Z})^2$ are the five Pythagorean families.*

2. *The Berggren child operators $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ and $C = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]$ are* parabolic *elements of $\mathrm{SL}_2(\mathbb{Z})$ ($\mathrm{tr} = 2$) fixing the cusps 0 and ∞ respectively. Their orbits generate the Barning–Berggren Pythagorean triple tree.*

3. *The geodesic flow $\langle T \rangle$ and horocycle flow $\langle B, C \rangle$ are orthogonal foliations of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.*

Proof. Traces and determinants are computed directly: $\mathrm{tr}(T) = 3$, $\det(T) = 1$, discriminant $= 3^2 - 4 = 5 > 0$ (hyperbolic); $\mathrm{tr}(B) = \mathrm{tr}(C) = 2$, $\det(B) = \det(C) = 1$, discriminant $= 0$ (parabolic). Fixed points of T satisfy $z^2 + z - 1 = 0$, giving $z = (-1 \pm \sqrt{5})/2 = 1/\varphi$ and $-\varphi$. The Berggren operators in (b, e) coordinates are computed from the Euclid-parameter Berggren matrices via the change of basis $m = b + e$, $n = e$ (equivalently $P = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]$), verified to give B and C as stated. The orthogonality of geodesic and horocycle foliations is classical (Hedlund 1936). □

Corollary 4 (Equidistribution across families)**.** *By Ratner's theorem [4], the horocycle flow on $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ is equidistributing. Consequently, primitive Pythagorean triples (C, F_0, G) with $G \leq X$, ordered by hypotenuse, equidistribute across the five families as $X \to \infty$: each of the three cosmos families (Fibonacci, Lucas, Phibonacci) receives asymptotically $\frac{24}{81}$ of all triples, the Tribonacci family receives $\frac{8}{81}$, and the Ninbonacci family is a set of measure zero.*

6 Discussion

Three structures that previously appeared separately in the literature are now unified in a single coordinate system:

Structure	Generator	Type
Five Pythagorean families	$T = F^2$ (QA step)	hyperbolic
Barning–Berggren tree	B, C (Berggren)	parabolic
Stern–Brocot embedding	F (Fibonacci step)	hyperbolic (GL_2)

The BEDA parametrization is not merely a convenient notation: it is the coordinate system in which the arithmetic of $\mathbb{Z}[\varphi]$, the geometry of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$, and the combinatorics of the triple tree are simultaneously legible.

Acknowledgments

[To be added]

References

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